Qn. Set Code-7(SOT)

Semester: 5th
Programme: B.Tech
Branch: Comp. Engg.

AUTUMN END SEMESTER EXAMINATION-2025

5rd Semester B.Tech

DATA MINING DATA WAREHOUSING

CS30013 / IT 3031

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Define supervised and unsupervised learning in data mining.

(b) Give the five-number summary of the data, where the age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(c) What is the importance of correlation analysis in understanding relationships between variables, and how can it support decision-making and predictive modeling?

(d) What is the role of entropy in decision tree algorithms?

(e) How many 3-item candidate sets are possible from 6 unique items?

(f) How many steps are there in the back propagation algorithm? What learning rule is used to implement gradient descent in back propagation networks?

(g) Compute the Jaccard similarity between the following two binary vectors:
X = {0011001100}, Y = {1110001110}

(h) Suppose the optimal separating hyperplane is given by $2x1 + 4x2 + x3 - 4 = 0$ and the class labels are +1 and –1. For the training example (1, 0.5, 1), what will be the class label and check is it a support vector?

(i) Draw flowchart of Genetic Algorithm.

(j) Given: TP=40, FP=10, FN=5, TN=45. Calculate F1-score.

SECTION-B

2. (a) Explain the concept of hierarchy in data warehousing. [5]
The Global Mart, a multinational retail company that sells products both online and in physical stores. The management wants to design a data warehouse to analyze business performance across sales, customers, products, and regions. Attributes:
Sales_ID, Product_ID, Customer_ID, Store_ID, Date_ID, Revenue, Quantity_Sold, Discount.

(Tables: Sales, Product, Customer, Store, Date)

Design three possible schema models (Star Schema, Snowflake Schema, Galaxy Schema)

i. Provide labeled diagrams for each.
ii. Explain the structure, characteristics, and use cases of each schema.

(b) Discuss at least two techniques that overcome the limitations of the Apriori algorithm. Consider the given transaction database. Find-out the frequent item sets and strong association rules with minimum support 50% and minimum confidence 75 % using FP-Growth tree. [5]

Transaction ID	Items Purchased
T1	Pen, Notebook, Stapler
T2	Pen, Notebook
T3	Pen, Stapler, Marker
T4	Notebook, Stapler, Envelope
T5	Pen, Notebook, Stapler, Envelope
T6	Notebook, Stapler

3. (a) Why k-NN is called as Lazzy Learner? Consider a set of five training examples given as $((x_i, y_i), c_i)$ values, where x_i and y_i are the two attribute values (positive integers) and c_i is the binary class label: {((1, 1), −1), ((1, 7), +1), ((3, 3), +1), ((5, 4), −1), ((2, 5), −1)}. Classify a test example at coordinates (3, 6) using a k-NN classifier with k = 3 and Manhattan distance defined by d((u, v), (p, q)) = |u − p| + |v − q|. [5]

(b) For the given set of objects, apply *k*-means clustering method to find the final two clusters and their centers assuming initial cluster centers to be A1 and A3. [5]

Objects	A1	A2	A3	A4	A5	A6
X1	2	2	8	5	7	6
X2	10	5	4	8	5	4

4. (a) What is Naive Bayes Classifier? Consider the following hypothetical data concerning student characteristics and whether or not each student should be hired. [5]

Name	GPA	Effort	Hirable
Sarah	poor	lots	Yes
Dana	average	some	No
Alex	average	some	No
Annie	average	lots	Yes
Emily	excellent	lots	Yes
Pete	excellent	lots	No
John	excellent	lots	No
Kathy	poor	some	No

Use a Naive Bayes classifier to determine whether or not someone with excellent attendance, poor GPA, and lots of effort should be hired.

(b) How to handle missing values in dataset? Discuss the different type of missing values with a suitable examples. [5]

5. (a) Predict the sales based on the expenditure on advertisement using simple linear regression. Calculate MSE and MAE. [5]

X (Advertising Spend in ₹ Lakhs)	Y (Sales in ₹ Lakhs)
1	1.8
2	2.4
3	3.2
4	3.8
5	4.5

(b) Why do we use DBSCAN? Apply DBSCAN clustering algorithm to cluster the following points with epsilon=1.5 and minimum points=2. [5]

Point	X	Y
A	1.0	1.0
B	1.2	1.1
C	0.8	1.2
D	5.0	5.0
E	9.0	9.0

6. (a) Predict whether a customer will subscribe to a mobile plan based on this following factors using Decision tree algorithm. [5]

I. Carry out the first split using information gain

II. Next level onwards carry out the decision tree construction using heuristic method.

III. Infer all the decision rules.

Customer	Age Group	Data Usage	Contract Type	Subscribed
1	Young	High	Prepaid	Yes
2	Young	Low	Postpaid	No
3	Middle	Medium	Prepaid	Yes
4	Senior	Low	Prepaid	No
5	Senior	High	Postpaid	Yes
6	Middle	High	Postpaid	Yes

(b) Cluster the following data points with agglomerative clustering with average linkage. [5]

Points	X	Y
A	2	3
B	3	4
C	5	8
D	6	9
E	8	10

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